CORA: deciphering the Swiss Federal Archives with Apertus

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Code and data: https://github.com/Anakalypto/CORA-Apertus · Demo video: link to be added · Licence: Apache-2.0

1. Summary

The Archive database of the Swiss Federal Archives describes 263,123 item-level documents in dense German, French and Italian cataloguing prose. It is rich but not AI-ready: encoding damage, labelled sub-fields packed into free text, dotted dates, author fields that mix roles, and no source typing or geography. **CORA** turns each record into AI-ready, provenance-tracked metadata with **Apertus v1.5**: a résumé of at most three sentences, document type and medium, a primary/secondary/tertiary source level with its rationale, Gallica/Dublin-Core-style metadata, persons, organisations and places, the preprocessing the record needs, and the document's story on a map. Rules, a gazetteer and Wikidata check and complete the model, and every field records where it came from. The output is browsable in a Gallica-inspired site with cited Q&A.

Measurement	Result
100-record demo (70B): records without structural or critical error	**99 %** (target ≥ 80 %); 1 % structural errors (target < 5 %)
Manual review, 25 demo records	**88 % usable** (32 % fully correct, 56 % minor errors), 12 % with a major error
10,000-record mixed batch (8B + 70B per feature)	**98.6 %** pass every automatic check; 0.9 % structural errors
Manual review, 66 batch records (all tiers, types, eras)	**62 / 66 without a major error**; 35 fully correct, 27 minor
Verification against the Bern datafication (Stapfer-Enquête 1799)	**90–100 %** agreement on 9 of 12 hand-coded fields
Scale run, Apertus 8B	**1,000 records in 48 min** (target 2 h); 0.4 % structural errors after repair
Apertus 8B vs 70B	8B as good for parsing and Q&A at 3× the speed; 70B needed for the source level

Table 1. Headline results. Sources: docs/quality_report.md, docs/batch10k_verification.md, docs/stapfer_verification.md, docs/model_comparison.md.

Our current open problem is the source level. The referral of records to primary, secondary and tertiary sources, both in the reference labels we built from the archive and in how Apertus is asked to decide, has to be reconsidered (§5).

2. System

The pipeline reads only the original archive-database fields (via LINDAS SPARQL, or the opendata.swiss bulk dump) and runs in independent stages, each reading the previous stage's file, so any stage can be rerun alone.

Stage	What happens	Model / tool
sample, fetch	Stratified random sample (seed 42) of rico:Record units; original fields only	LINDAS SPARQL
classify	Pass 1: résumé, type, medium, metadata, entities, story events (strict JSON schema, controlled vocabularies)	Apertus 8B
audit	Pass 2: eleven yes/no checks, each with quoted evidence; preprocessing flags derived by rule, OR-ed with deterministic detectors	Apertus 8B + rules
level	Pass 3: primary / secondary / tertiary with rationale; press convention configurable	Apertus 70B
repair, enrich	ISO dates, control characters; gazetteer, Wikidata links with life-date check, story trail to Bern	rules, Nominatim, Wikidata
qc, export	Automatic checks, edit log, JSONL/CSV/TSV + JSON schema, SQLite search index	rules

Table 2. Pipeline. Each field carries its provenance: record, derived, apertus, gazetteer or wikidata; conflicts are settled by fixed, logged rules (e.g. the accession year always comes from the accession number, coordinates always from the gazetteer).

Use of Apertus. Apertus v1.5 runs on CSCS through the Keera gateway (OpenAI-compatible; *LLM_NAME*, *LLM_BASE_URL*, *LLM_API_KEY*), temperature 0.1, JSON mode, up to 3 retries with the parse error fed back. Each feature runs on the model that won the head-to-head comparison (§3.3), configurable per feature in *src/config.toml*. Prompts carry strict grounding rules (never invent names, dates or places) and archival definitions. Asking the model directly for the six preprocessing flags returned “yes” for nearly every record; replacing it with

evidence-quoting checks (pass 2) made the flags informative.

Applications on the dataset. *Ask the archive* answers only from retrieved records (BM25 / SQLite index) and cites every sentence; citations to records that were not retrieved are dropped. A read-only **MCP server** exposes search, record and Q&A. Record pages show a **trust card** separating evidence (original fields) from context (Wikidata, Wikipedia, HLS). Further views: federal consultations 1960–1991 linked to Swissvotes, Parliament since 1848, a backtested referendum model, and **Swiss Tales**: a timeline of Swiss history with the archive units of each year, and *Oral tradition*, a cited, explicitly imagined year in the life of an ancestor from the 1799 Stapfer survey, censuses and the Federal Gazette, where Apertus also checks whether the place of origin existed and belonged to the Confederation in that year.

3. Evaluation

3.1 Quality of the classification

Demo, 100 records (70B). All 100 outputs were valid JSON with no retry; 99 % have no structural or critical error; résumés are within three sentences in 100 %; 92 % of the 377 named entities occur literally in the record and all 30 others are translations (Nationalrat → National Council), not inventions; creation dates fall inside the record's own range in 100 %. Apertus is weak on the accession date (24 % right), so a rule replaced it.

Manual review. On 25 demo records, 88 % are usable. All three major errors come from archival conventions the model does not know: speaker ≠ author, depositing agency ≠ creator, *Postulat* ≠ motion. On the 10,000-record batch we reviewed two records per tier × type × era cell (66 records):

Field	Title	Creator / publisher	Dates	Entities	Places	Personal data	Document type	Résumé	Source level
Accuracy	95.5 %	95.5 %	95.4 %	93.9 %	97.0 %	97.0 %	87.9 %	84.6 %	92.4 %

Table 3. Field accuracy in the manual review of the 10,000-record batch (66 records). The four major errors: a misread institution (German BND for the Swiss intelligence service), an invented first name, a grant request read as the book it funds, and a police card with the archive as creator.

3.2 Verification against the Bern case study

Apertus 70B read only the transcribed answers of random Stapfer-Enquête response sheets (1799) and extracted the 12 fields that the University of Bern team coded by hand. With the Bern codebook in the prompt and names compared across historical spellings (“Tachsen” = Dachsen), agreement is 98 % on the teacher's age, 100 % on the teacher's name, 95 % on school type, marital status and place, 94 % on winter pupils, 91 % on years of service and 90 % on confession. The naive prompt reached 0 % on school type and 26 % on confession: Bern's coded categories are conventions, and giving the model the codebook is what closes the gap. Remaining errors are mostly historical spellings and roles given instead of names.

3.3 Apertus 8B vs 70B

Every AI feature was run on both models with identical inputs, prompts, post-processing and scoring. Where a reference exists we score against it; otherwise we only compare the two models.

Feature (reference)	70B	8B	Choice
Parsing, 200 new records (automatic checks)	98.5 % pass	**100 %** pass	8B
– creator = record's author field	86.6 %	**95.1 %**	
Stapfer extraction, 100 sheets (Bern hand coding)	**84.5 %**	80.8 %; 8 % unusable	70B
Source level, 29 adjudicated records	**62 %**	55 %	70B
– secondary sources recognised (n = 120)	**70 %**	28 %	
– tertiary sources recognised (n = 110)	**93.6 %**	77.3 %	
Ask the archive, 24 questions: right record cited / unanswerable refused	100 % / 8 of 8	100 % / 8 of 8 (1 flag missing)	8B
“See more” explanations, 30 records: unsourced sentences	**2.5 %**	29.3 %	70B
Consultation tagging, 100 procedures: off-list topics	**3**	18	70B
What-if party positions, 40 real votes (naive baseline 71.7 %)	65.0 %	61.4 %	neither beats the baseline
Mean seconds per call (parsing / audit / level / Q&A)	16.5 / 6.9 / 3.3 / 2.4	**5.7 / 2.4 / 0.9 / 0.8**	8B ≈ 3× faster

Table 4. Apertus v1.5 70B vs 8B (docs/model_comparison.md, data/model_eval/). Several references are small (29 levels, 24 questions): differences of a few points are within noise.

Reading. 8B is enough for the passes a reader meets on every record (parsing, audit, Q&A) and is about three times faster, which matters for an archive that wants to host the model on a single GPU. 70B is still worth it where judgement or long, messy input dominate: the source level (8B misses most secondary sources), long transcriptions (8B copies them into its evidence field and breaks the JSON), and explanations that must stay sourced. Rules close several 8B gaps cheaply: mechanical detectors, a topic synonym map, and mapping “yes / free vote” to codes. Neither model beats a naive baseline on predicting party positions, so the what-if view stays labelled as imagination, not as a forecast.

3.4 Scale and cost

Run	Records	Wall-clock	Errors
Demo, Apertus 70B	100	≈ 9 min (325k prompt + 124k completion tokens)	1 % structural
Scale run, Apertus 8B, 12 parallel requests	1,000	**48 min**: model 15 min, throttled geocoding 32 min	6.0 % → 0.4 % after rule repair
Mixed batch, 8B (classify, audit) + 70B (level)	10,000	classify 83, level 54, audit 35, enrich 58 min (23.2M + 7.6M tokens)	0.9 % structural, 0.6 % critical

Table 5. Throughput. Two-thirds of the 1,000-record run is the public geocoder's 1 request/s limit; a self-hosted Nominatim removes it.

The 10,000-record batch was drawn over three source tiers (primary by form; secondary and tertiary from archival evidence: title conventions and form), three form types and five eras, none already in the demo. 98.6 % pass every automatic check, creation dates fall inside the record's range in 99.3 %, and 98.7 % of named entities are found in the record.

4. Deployment and sovereignty

- **On-premise and sovereign cloud.** The only runtime dependency is an OpenAI-compatible Apertus endpoint: the CSCS one, or Apertus weights served with vLLM on the archive's own hardware (8B fits on one GPU). Python (pandas, FastAPI) and a static front end with vendored MapLibre.
- **Air-gapped.** Serve Apertus locally, self-host Nominatim or ship the enrichment cache, serve swisstopo tiles locally and load records from the opendata.swiss bulk dump instead of LINDAS. Once built, the demo data runs offline except for map tiles.
- **Transparency.** Every value shows its provenance; every Apertus answer that a reader sees is cited or marked as imagined; raw outputs of both models are kept for audit.

5. Current problem: the source level must be reconsidered

Primary, secondary and tertiary seem simple, but in an administrative archive they are not, and this is where our results are least solid. Two things need to be reconsidered: **how the datasets refer records to a level** (our reference labels) and **how Apertus is asked to decide**.

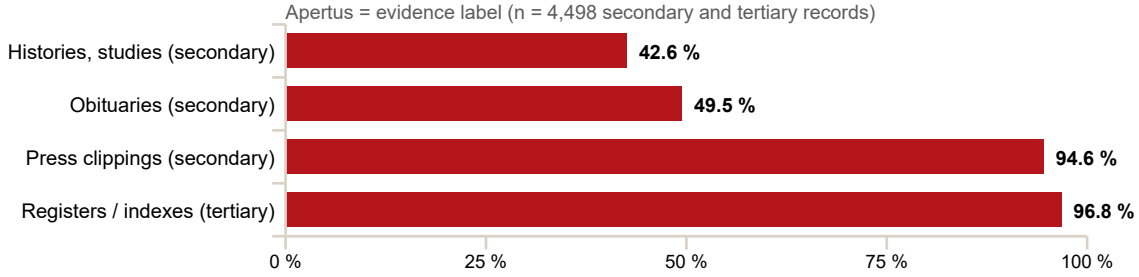


Figure 1. Source level in the 10,000-record batch: agreement of Apertus (70B) with the level implied by archival evidence. Overall 78.7 %, but very uneven by stratum.

- **The reference labels are unreliable.** We derived secondary and tertiary labels from title conventions and form (“Register”, “Verzeichnis”, “Studie”, clipping dossiers). On 200 such units Apertus agreed only 39.5 % of the time, yet when we adjudicated 30 disagreements by hand, **Apertus was right in 19, the label in 4**, and 6 were defensible both ways. A “Verzeichnis” in a federal archive is mostly a primary list of firms, material or persons, not a finding aid; a “Studie” is often a contemporary planning study.
- **The same holds at scale.** In the batch, registers and press clippings agree 95–97 %, but “history” and “obituary” titles only 43–50 %: those patterns also catch department dossiers about congresses, correspondence about a Festschrift and parliamentary eulogies, which Apertus correctly calls primary. Judged record by record in the manual review, the level is right in 92 %.
- **Conventions are inconsistent.** Press is secondary as a second-hand account, but historians use it as a primary source for public discourse. Our press convention was applied to broadcast news only part of the time (1,133 of 2,248 TV news and newsreel records in the batch).

- **Model gaps remain.** 70B recognises 70 % of secondary sources (8B: 28 %), misses some true indexes, and once called a published 1881 treatise primary because it was “written at the time”.

What we will change. (1) Replace keyword labels with a gold set adjudicated by archivists, stratified like the batch. (2) Give the prompt explicit archival definitions: *tertiary* = a finding aid, index or register that points to other records; a list of persons, goods or places is primary; *secondary* = an account made after the events by someone not involved. (3) Separate the level from the genre: a press/broadcast flag, with one documented convention applied by rule to all news, and the level shown as relative to the research question. (4) Report the level with its rationale and confidence and let users correct it through the edit log. (5) Fine-tune 8B on the adjudicated set if 70B remains necessary.

6. Limitations and ethics

- **Content accuracy.** The weakest fields are the résumé (84.6 % in the batch review) and document type on non-standard items (posters, typed communiqués). Apertus sometimes adds first names from world knowledge; a Wikidata life-date check catches wrong links.
- **Blind spots covered by rules.** Apertus does not notice encoding damage (recall 0 of 38) and rarely fills the accession date; 8B writes more non-ISO dates and sometimes control characters in place of accents. Deterministic detectors and repairs handle these.
- **Personal data.** The anonymisation flag has about 40 % precision (it confuses sensitive topics with sensitive persons): a triage aid, not a decision. Nothing personal is added beyond Wikidata links to public figures; Oral tradition keeps family stories in the browser and only uses archive units older than 1950. See docs/ethics_impact_assessment.md.
- **Small references.** Manual reviews were done by our team on 25 + 66 records; archivists of the Federal Archives should repeat them on a larger sample.

7. Reproducibility

`pip install -r requirements.txt; cp .env.example .env (set LLM_API_KEY); python src/run.py all; python src/run.py serve`. Each stage can be rerun alone (*python src/run.py qc*); the 10,000-record batch is *src/batch_run.sh*, the model comparison *src/eval_features.py*, the Bern verification *python src/run.py stapfer*. Sample seeds are fixed (42, 1000, 7); model temperature is 0.1, so reruns vary slightly. Python 3.12, Apertus v1.5 on vLLM, LINDAS data of 2026-01-08. A demonstration notebook is in notebooks/demo.ipynb.

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Repository documents: docs/quality_report.md, docs/model_comparison.md, docs/batch10k_verification.md, docs/stapfer_verification.md, docs/ethics_impact_assessment.md.